EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

OPERATION PALE HORSE

ARM Cortex-M33 • GDB • Ghidra

November 14, 2026

For educational and defensive security research only

A Story About a Very Strange Little Computer

1. The Box

A worker at a shipping warehouse near a harbor opened a package that had sat unclaimed for months.

Inside was a small computer.

It was not a laptop. It was not a phone. It was a tiny circuit board, about the size of a stick of gum, with wires soldered on by hand. Three small lights. One button. A little connector for a cable. A dead watch battery.

There was no brand name. No serial number. No instructions. Nothing to say who made it or what it was for.

It looked like something a student might build for a science fair.

It was not.

2. The Strange Message

An investigator plugged the board into a computer and opened a program to watch the messages coming out of it.

At first, nothing happened. Then eight letters appeared:

```
PALE
```

Then the board went silent. It would not answer. It would not print anything else. It would not even say hello.

The investigator tried again and again. Every time, the same eight letters, then silence.

That was the first clue. A normal computer tells you what it is doing. This one was refusing to talk.

3. The Company That Is Not Really a Company

A search of the shipping records found one name attached to the package:

Fun Society Amusement, LLC

It sounds like a company that runs an arcade. And there really is an arcade, in a small town, with neon signs and row after row of games. It pays its taxes. It has a website. It even sponsors a kids' go-kart team.

But the company has never made a profit. It has never tried to. The arcade is a front. It is a fake business that hides a real one.

And the little board in the evidence bag was built by the real one.

4. What the Board Really Is

The board is a **dead drop**.

A dead drop is a hiding place where spies leave messages for each other without ever meeting. One person hides a message. Later, another person picks it up. Nobody sees them together.

This board is a digital dead drop. It holds a secret list. When the right person arrives and proves who they are, the board gives them the list. Anyone else gets nothing.

The list is called the **manifest**. It holds the names of every place and every person in the network. It is the one thing that can shut the whole operation down.

The board is protecting it with three separate locks.

---

5. The First Lock: The ID Check

Before the board will run its main program, it does an ID check.

It takes a fingerprint of the program stored inside it. Then it compares that fingerprint to the one it remembers. If the two match, the board starts up. If they do not match, the board simply does not start.

This means you cannot change the program. The moment you do, the fingerprint changes, and the board goes quiet.

Think of it like a bouncer at a door who checks your ID against a list. If your name is not on the list, you do not get in. No argument, no explanation. You just do not get in.

---

6. The Second Lock: The Hidden Room

The most important program inside the board is not stored where you can see it.

It is stored **scrambled**, in a form that looks like random noise. When the board starts up, it unscrambles that program, copies it into temporary memory, and runs it from there.

Temporary memory is called **RAM**. RAM does not keep anything when the power goes off. So if you take the board apart and look at its permanent memory, you will never find the real program. You will only find the scrambled copy.

Think of it like a secret room with no door. The only way in is a hidden passage that exists for a few seconds at a time.

---

7. The Third Lock: The Password System

The board listens on a cable. You can type commands to it.

But the board is clever. If you type `help`, it shows you a list of commands. That list is not complete. There is a hidden command called `factory` that gives you the manifest, and the board never mentions it.

There is also a password. The board gives you a number, and you have to send back the right answer. The number changes every single time, so you cannot just copy an answer from before. You have to understand the rule that creates the answers.

Think of it like a riddle that is different every time you knock. You cannot memorize the answer. You have to know how the riddle works.

8. The Danger

The board has one more trick. If it thinks someone is poking at it the wrong way, it shuts itself down and refuses to work.

For a long time, investigators thought this meant the board could destroy the manifest if they made a mistake.

That would be the worst possible outcome. The manifest is the only copy. If the board wipes it, the network wins.

So every step has to be careful. You do not smash the box open. You do not rush. You watch it, you learn it, and then you open it the way it was meant to be opened.

---

9. The Good News

Here is the part that makes this possible.

The people who built the board were clever, but they made one mistake. They kept their plans. The source code, the instructions for building the board, was left on a computer the network thought was safe.

Because of that mistake, you do not have to guess. You can read the plans. You can build the board yourself. You can watch it run. And then you can take it apart, piece by piece, until you understand every part of it.

That is not cheating. That is how real reverse engineers learn. First you study a machine you understand. Then you can take apart a machine you do not.

---

10. Your Mission

You have the board. You have a cable to watch what it says. You have a tool to watch it run, and a tool to read its insides.

Your mission is simple to say and hard to do:

1. Read the plans and build the board.
2. Watch the board start up, one step at a time.
3. Find the fingerprint check and learn how it works.
4. Follow the hidden program into temporary memory.
5. Find the hidden command and the password rule.
6. Read the manifest.

You will do all of this without breaking the board, and you will understand every single step, because by the end you will have built it yourself and taken it apart with your own hands.

The manifest is in there. It has a company name on it.

Go get it.

The One Sentence Version

A fake arcade company built a tiny computer to hide a secret list behind three locks, and your job is to open all three without breaking it.